



Course Name

GENERAL PRACTICAL PHYSICS II

Topic: Young's Modulus

WEEK 12

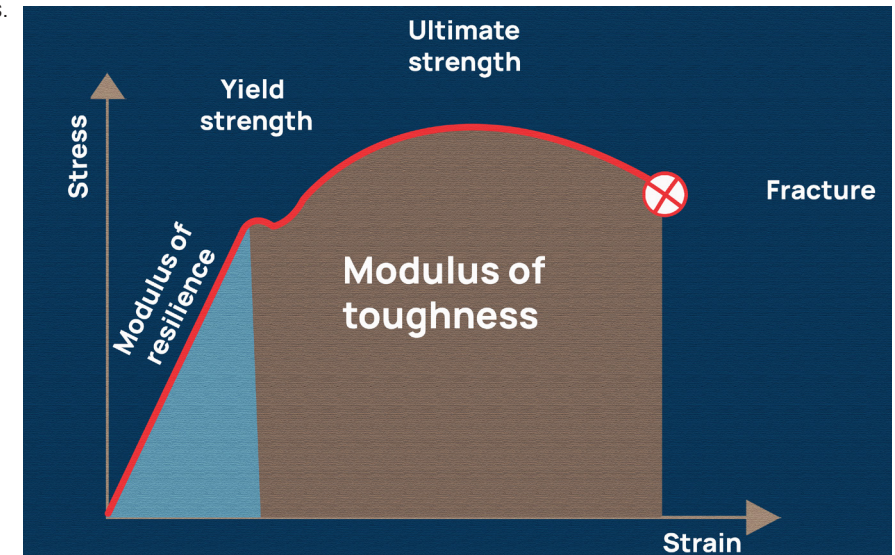
IN SUMMARY

This week, I learned about Young's Modulus, a fundamental property of materials that measures their stiffness. It helps us understand how materials respond to stretching or compression. Young's Modulus can be calculated using the formula $Y = \text{stress}/\text{strain}$, where stress is the force applied to a material divided by its cross-sectional area, and strain is the change in length compared to the original length.

Young's Modulus is important in many engineering applications. It helps engineers design structures that can withstand the forces they will encounter. For example, Young's Modulus is used to design suspension bridges, airplane wings, mattress springs, and automobile suspension systems.

BULLET POINT SUMMARY

- Young's Modulus is a fundamental property of materials that measures their stiffness.
- Young's Modulus can be calculated using the formula $Y = \text{stress}/\text{strain}$.
- Young's Modulus is important in many engineering applications, such as designing suspension bridges, airplane wings, mattress springs, and automobile suspension systems.
- Other applications of Young's Modulus include biomechanical applications, such as designing prosthetic limbs, orthopaedic implants, and athletic shoes.
- Young's Modulus is also used in the development of high-tech materials for aerospace, defence, and electronics industries.



CASE STUDY

Title: Using Young's Modulus to Design More Efficient Aircraft Wings

Introduction

Airplane wings are designed to withstand aerodynamic forces and turbulence during flight. Young's Modulus is a measure of the stiffness of a material, and it is important to select materials with a high Young's Modulus for airplane wings. This ensures that the wings will return to their original shape after bending, which is critical for the structural integrity and safety of the aircraft.

For example, engineers at Boeing used Young's Modulus to design the wings of the Boeing 787 Dreamliner. The wings of the 787 Dreamliner are made of composite materials, which have a higher Young's Modulus than traditional materials such as aluminum. This allowed Boeing to design wings that are lighter and more efficient than the wings of previous aircraft.

CASE STUDY

Objective

The objective of this case study is to demonstrate how Young's Modulus can be used to design more efficient aircraft wings.

Methodology

Engineers use a variety of methods to determine the Young's Modulus of different materials. One common method is to perform a tensile test. In a tensile test, a sample of the material is pulled and the force required to stretch the material is measured. The Young's Modulus can then be calculated using the formula $Y = \text{stress}/\text{strain}$.

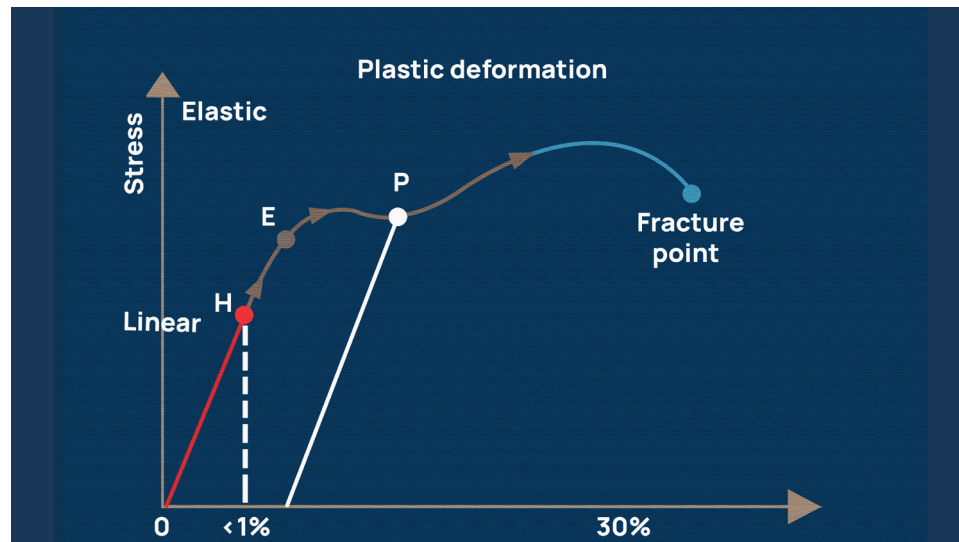
Once the Young's Modulus of different materials is known, engineers can select the best material for a particular application based on the desired stiffness and other factors such as weight and cost.

Results

By using Young's Modulus to design aircraft wings, engineers can create wings that are lighter and more efficient. This can lead to reduced fuel consumption and increased range for aircraft.

QUESTIONS TO PONDER

- How can Young's Modulus be used to design more efficient aircraft wings?
- What are some other challenges that engineers face when designing aircraft wings?



SKILLS AND COMPETENCIES YOU HAVE ACQUIRED AFTER THIS LESSON

Ability to define Young's Modulus and explain its significance.

Ability to calculate Young's Modulus using experimental data.

Ability to communicate the results of Young's Modulus experiments and analyses in a clear and concise manner.

PERSONAL REFLECTION

I learned a lot about the fundamental concepts of Young's Modulus and its wide range of applications. I was particularly interested in the case study on using Young's Modulus to design safer airplane wings. It is amazing to think that such a simple concept can have such a profound impact on the safety and efficiency of modern aircraft.